

MOLLUSCUM CONTAGIOSUM VIRUS DETECTION BY REAL-TIME PCR

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Introduction

Molluscum contagiosum virus (MCV) is a member of the poxvirus family, that causes skin lesions. MCV and variola are the only pox viruses that solely infect humans. It is a double-stranded DNA virus of approximately 180kpb in size, and is distributed worldwide. MCV generally remains localized at the site of the lesions, however immuno-compromised people may have lesions that persist for many months to years. Diagnosis is commonly based on the appearance of the lesions, however a differential diagnosis may be required. A rapid real-time TaqMan assay for MCV has been developed to aid in the diagnosis of this virus.



Specificity

The MCV TaqMan assay did not detect the following viruses.

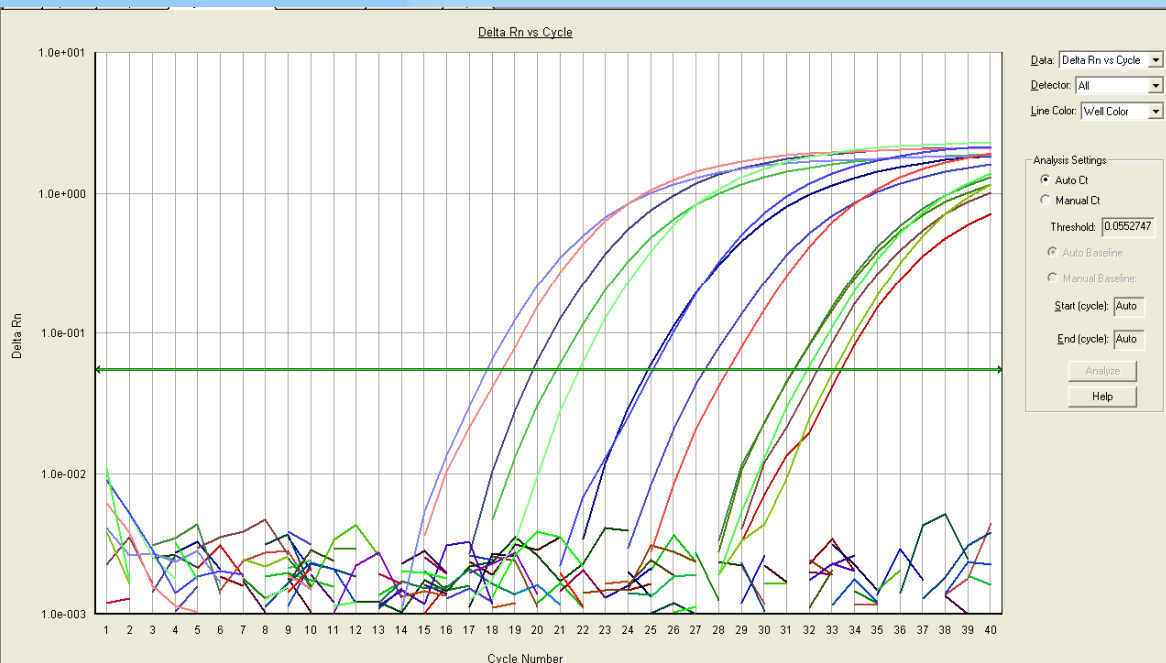
- ❖Bovine herpes virus (BHV)
- ❖Parapox virus
- ❖Varicella zoster virus (VZV)
- ❖Cowpox virus
- ❖Herpes simplex virus (HSV)
- ❖Tanapox virus
- ❖Measles virus
- ❖Monkeypox virus
- ❖Vaccinia

Sensitivity

Ten fold dilutions of DNA were prepared and tested in parallel with the existing single round PCR for MCV. Both the TaqMan and the PCR were performed at the same time. The PCR detected the patient DNA at 10⁻⁴. The p43K TaqMan detected the same patient DNA at 10⁻⁶ with a ct value of 36.89. The TaqMan method was considered more sensitive.

Synthetic Controls

Two synthetic controls were developed for this test (data not shown). One is specifically designed as a control of the probe, the other as a control of the primers in the assay. The rationale for this is two fold. Firstly, for troubleshooting purposes it allows for rapid pinpointing of either the probe or the primer failure. Secondly, a consistent volume of material is available so that run-to-run variation can easily be monitored, without relying on finite patient material.



Method

Reagents

Probes and primers were designed using sequences of the p43k gene available on Genbank. Together with Applied Biosystems TaqMan reagents, a mix was formulated and optimized for use. The mix consisted of 1x TaqMan Fast Universal PCR mix with 150nM probe and 300nM primers.

Cycling conditions

50°C for 5min*, 95°C for 2min, followed by 40 cycles of 95°C for 3sec and 60°C for 30sec.
*50°C step can be omitted if RNA tests are not simultaneously amplified with this test.

Probe and Primers

Name	Sequence (5'- 3')
MCV p43K F	GCTCACGTACGACTGCTTYGAC
MCV p43K R	CGTGGAGCGCAGATTGC
MCV p43K P	6FAM-CGCTCATCTCGCAGAC -MGB

Results

Samples

DNA from samples submitted for MCV PCR between 2004 and 2009 were tested using the new rapid real-time TaqMan assay. The DNA was extracted using various Qiagen methods over this time, including the QIAamp DNA Blood Mini Kit , EZ1 Virus Mini Kit , and the QIAamp Virus BioRobot MDx Kit . A total number of 74 swab samples were tested. From these samples, 40 had been MCV detected by PCR on submission. The new real-time TaqMan test detected 49 samples with ct values between 21-34 cycles. The nine extra MCV detected samples, were submitted to a sequencing PCR (data not shown), with MCV confirmed in every sample.

Discussion and Conclusion

The sensitivity and specificity of this real-time TaqMan assay for MCV achieves a better testing regime in the laboratory. The method is fast, reliable, and specific and has many advantages over performing a gel-based PCR in a routine diagnostic laboratory.

Acknowledgement

All the members of the Molecular Virology team who have diligently tested patient samples for MCV since 2004.